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I) FILE OPERATIONS

```
vboxuser@Ubuntu: ~/prathana/batch31ai/linux_lab
vboxuser@Ubuntu:~/prathana/batch31ai$ ls
file.txt
vboxuser@Ubuntu:~/prathana/batch31ai$ ~/linux_lab
bash: /home/vboxuser/linux_lab: No such file or directory
vboxuser@Ubuntu:~/prathana/batch31ai$ ^C
vboxuser@Ubuntu:~/prathana/batch31ai$ cd ~/linux_lab
bash: cd: /home/vboxuser/linux_lab: No such file or directory
vboxuser@Ubuntu:~/prathana/batch31ai$ pwd
/home/vboxuser/prathana/batch31ai
vboxuser@Ubuntu:~/prathana/batch31ai$ mkdir linux_lab
vboxuser@Ubuntu:~/prathana/batch31ai$ cd linux_lab
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ pwd
/home/vboxuser/prathana/batch31ai/linux_lab
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ nano greeting.txt
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ cat greeting.txt
Hello Linux

vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ cp greeting.txt backup.txt
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ ls
backup.txt  greeting.txt
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ mv backup.txt old_greeting.txt
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ ls
greeting.txt  old_greeting.txt
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ mkdir temp_dir
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ rmdir temp_dir
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ ls -l greeting.txt
greeting.txt
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ ls -l greeting.txt
-rw-rw-r-- 1 vboxuser vboxuser 13 Sep  8 05:44 greeting.txt
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ ^C
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$
```

Fig 1: Essential File Commands

Summary

- Set up a working directory called `linux_lab` with the `mkdir` command and entered it using `cd`.
- Created a file named `greeting.txt` through `nano` and wrote "Hello Linux" inside.
- Displayed and checked the text stored in the file using `cat`.
- Made a copy of `greeting.txt` as `backup.txt` with the `cp` command.
- Changed the name of `backup.txt` to `old_greeting.txt` using `mv`.
- Created a temporary folder called `temp_dir` with `mkdir` and removed it afterward using `rmdir`.
- Used `ls -l` to view the permissions, ownership, size, and other details of `greeting.txt`.

II) APT PACKAGE MANAGEMENT

```
vboxuser@Ubuntu: ~/prathana/batch31ai/linux_lab
~/prathana/batch31ai/linux_lab

vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://np.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://np.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:5 http://np.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Get:7 http://np.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [601 kB]
Get:8 http://np.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.3 kB]
Get:9 http://np.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [409 kB]
Get:10 http://np.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [79.5 kB]
Get:11 http://np.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [268 kB]
Get:12 http://np.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:13 http://np.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.6 kB]
Get:14 http://np.archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3,036 B]
Get:15 http://np.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1,056 B]
Fetched 2,161 kB in 3s (652 kB/s)
9 packages can be upgraded. Run 'apt list --upgradable' to see them.
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$
```

Fig 2: Updating Ubuntu Package Lists Using `sudo apt update`

```
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ sudo apt install build-essential
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30      linux-image-unsigned-7.0.0-30-generic  linux-modules-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  build-essential

Installing dependencies:
  bzip2      g++-15-x86-64-linux-gnu  gcc-x86-64-linux-gnu  libcc1-0
  dpkg-dev  g++-x86-64-linux-gnu    libalgorithm-diff-perl  libcrypt-dev
  fakeroot  gcc                    libalgorithm-diff-xs-perl  libdpkg-perl
  g++       gcc-15                 libalgorithm-merge-perl  libfakeroot
  g++-15   gcc-15-x86-64-linux-gnu  libasan8                libfile-fcntllock-perl

Suggested packages:
  bzip2-doc      g++-multilib  gcc-multilib  libtool  gcc-doc  g
  debian-keyring  g++-15-multilib  autoconf    flex    gcc-15-multilib  g
  debian-tag2upload-keyring  gcc-15-doc    automake    bison    gcc-15-locales  b

Summary:
  Upgrading: 0, Installing: 31, Removing: 0, Not Upgrading: 9
  Download size: 58.8 MB
  Space needed: 205 MB / 42.8 GB available
```

Fig 3: Installing the Build-Essential Package Using `sudo apt install build-essential`

```
~/prathana/bat
vboxuser@Ubuntu:~/prathana/batch31ai$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

vboxuser@Ubuntu:~/prathana/batch31ai$
```

Figure 4: Checking the Installed GCC Compiler Version Using `gcc --version`

```
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ apt search python3
alembic/resolute 1.18.4-1 all
  lightweight database migration tool for SQLAlchemy

antimony/resolute 0.9.3-2.1build2 amd64
  Computer-aided design CAD tool

astro-frameworks/resolute 5.0 all
  Astronomical data analysis frameworks

astro-python3/resolute 5.0 all
  Python 3 packages for astronomy

bamkit/resolute 0.0.1+git20170413.ccd079d-3build1 all
  tools for common BAM file manipulations

beancount/resolute 3.2.0-2build1 all
  Double-entry accounting from text files

berrynet-dashboard/resolute 3.10.2-2build1 all
  deep learning gateway - python3 modules

brltty/resolute,now 6.7-1ubuntu6 amd64 [installed,automatic]
  Access software for a blind person using a braille display

budgie-app-launcher-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
  Applet to provide an alternative means to launch applications

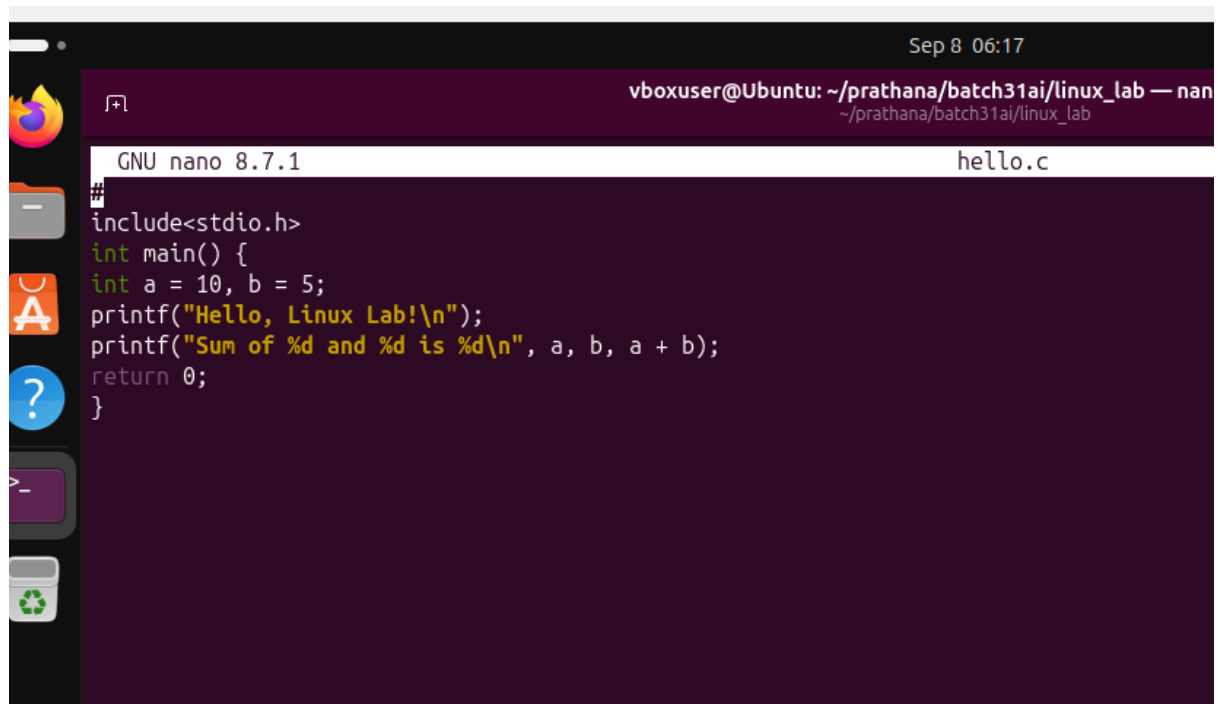
budgie-brightness-controller-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
  Applet to control the brightness of the screen
```

Figure 5: Searching for Python 3 Packages Using `apt search python3`

Summary

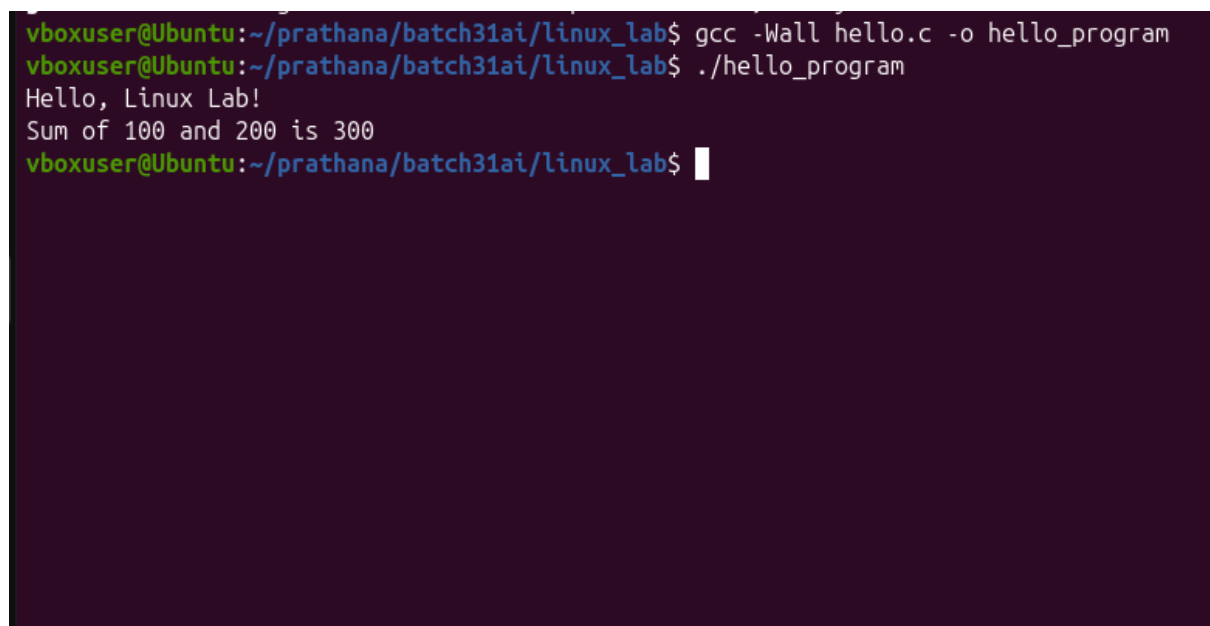
- Updated the Ubuntu package lists using the `sudo apt update` command to retrieve the latest available package information.
- Installed the build-essential package with `sudo apt install build-essential`, including the required compiler and development dependencies.
- Checked the installed GCC compiler version using `gcc --version` and confirmed that GCC was successfully available on the system.
- Used `apt search python3` to search the Ubuntu repositories for packages related to Python 3 and view the available options.

III) C COMPILATION



```
GNU nano 8.7.1 hello.c
#include<stdio.h>
int main() {
int a = 10, b = 5;
printf("Hello, Linux Lab!\n");
printf("Sum of %d and %d is %d\n", a, b, a + b);
return 0;
}
```

Figure 6: Creating the C Program hello.c Using the Nano Editor



```
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ gcc -Wall hello.c -o hello_program
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 100 and 200 is 300
vboxuser@Ubuntu:~/prathana/batch31ai/linux_lab$
```

Figure 7: Compiling and Executing the C Program Using GCC

Summary

- Created a C source file named `hello.c` using the nano text editor and added a simple program to display a greeting and calculate the sum of two numbers.
- Compiled the C program using `gcc -Wall hello.c -o hello_program` to generate an executable file.
- Ran the compiled program using `./hello_program` and successfully displayed "Hello, Linux Lab!" along with the calculated sum of 100 and 200 as 300.

IV. CLEANUP AND VERIFICATION

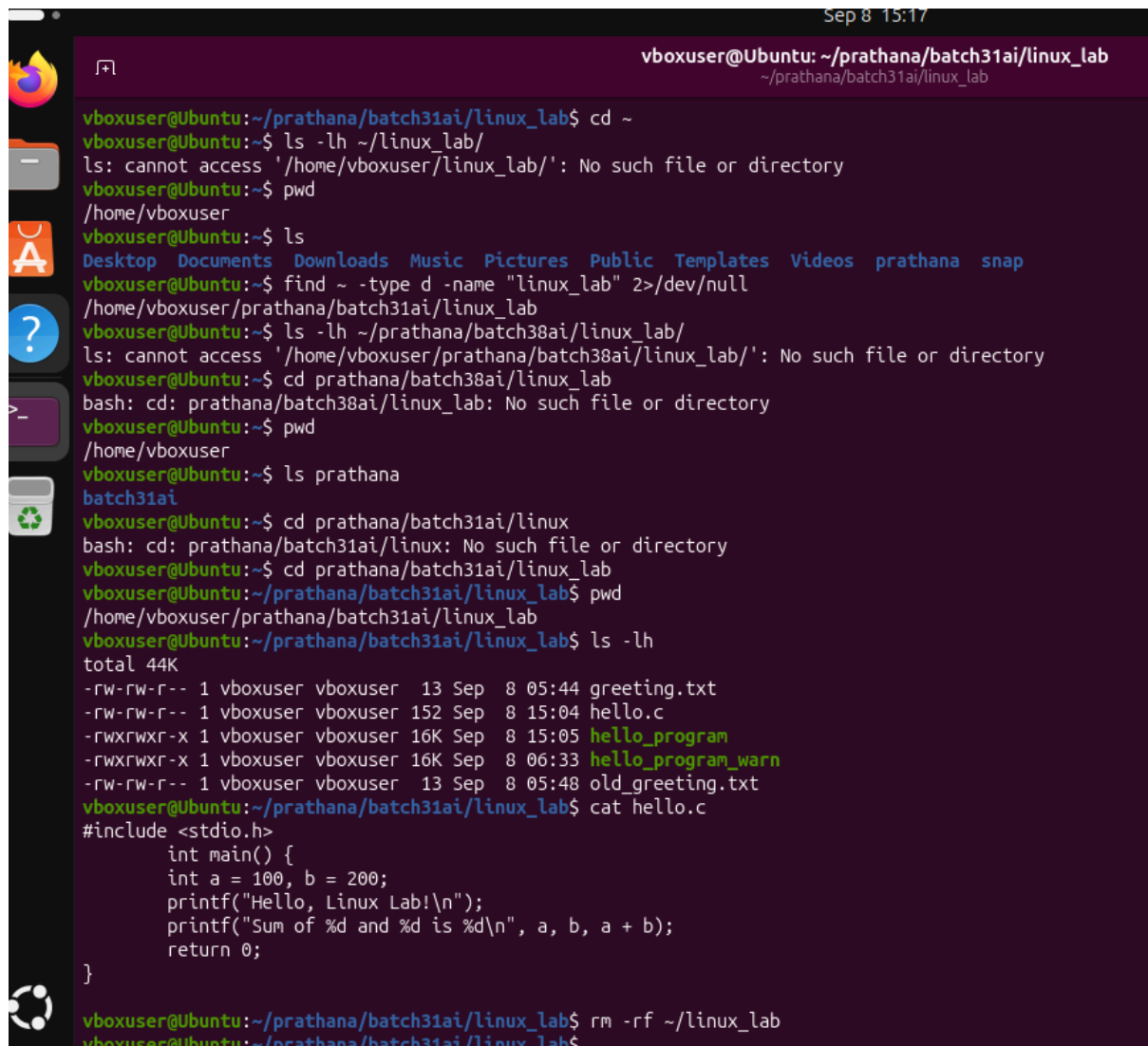
A terminal window titled 'vboxuser@Ubuntu: ~/prathana/batch31ai/linux_lab' with a timestamp 'Sep 8 15:17'. The terminal shows a series of commands and their outputs. The user starts by navigating to their home directory and attempting to list files in ~/linux_lab, which fails. They then check their current directory and list files in the home directory. Next, they use 'find' to locate a directory named 'linux_lab' and attempt to navigate into it, failing. They then try to navigate into a subdirectory 'prathana/batch38ai/linux_lab', also failing. After returning to the home directory, they list the contents of 'prathana', showing 'batch31ai'. They then navigate into 'prathana/batch31ai/linux' and attempt to navigate into 'linux_lab', failing. Finally, they navigate into 'prathana/batch31ai/linux_lab' and list its contents, showing files like 'greeting.txt', 'hello.c', 'hello_program', 'hello_program_warn', and 'old_greeting.txt'. They view the contents of 'hello.c' using 'cat', showing a C program that prints a greeting and calculates the sum of two numbers. The session ends with the user attempting to remove the ~/linux_lab directory.

Figure 8: Creating the hello.c Source File Using Nano

Summary

- Created the hello.c file in the nano editor and wrote a C program to print a greeting and calculate the sum of two numbers.
- Compiled the program with GCC using the gcc -Wall command and created the executable hello_program.
- Executed the compiled program with ./hello_program and successfully displayed the greeting and the sum 300.